ZCAS UNIVERSITY

BSCSE32 - BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

CCS3241 SOFTWARE QUALITY ASSURANCE AND TESTING

FINAL EXAMINATIONS

MONDAY, 30th MAY, 2025

TIME: 14:00 – 17:00 HOURS

TIME ALLOWED: THREE (03) HOURS

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is **compulsory** with **ONE (01)** Question and carries **40 marks.**
2. **SECTION B:** Attempt **any THREE (03)** questions from this section. Each question carries **20 marks**.
3. This examination paper carries a total of **100 marks.**
4. Time allowed: **THREE (03) Hours** plus **FIVE (05) minutes** to read through the paper.
5. **DO NOT** turn this page **until** you are told to do so.

SECTION A – COMPULSORY QUESTIONS

INSTRUCTIONS: You MUST attempt **ALL QUESTIONS** that follow the case study provided.

QUESTION ONE

CASE STORY: "Mankwara Medicines Regulatory Authority (MMRA)"

The **Mankwara Medicines Regulatory Authority (MMRA)** is a public institution that has embarked on a digital transformation project to digitize drug approval, pharmacovigilance, and inspection records. However, recent audits revealed weaknesses in cybersecurity practices across departments. Systems lacked documented security procedures, had inconsistent access control mechanisms, and were often updated manually.

To address this, **MMRA** has decided to adopt the **Capability Maturity Model (CMM)** as a framework to assess and improve its cybersecurity posture. The ICT unit is tasked with performing an initial capability maturity assessment and designing a roadmap toward a fully optimized and measurable cybersecurity program.

Tasks:

a) Describe the **Capability Maturity Model (CMM)** and explain its purpose in evaluating software development processes in public institutions. **(06 marks)**

b) Describe the **(05) five** levels of maturity in the CMM, providing relevant cybersecurity examples for each level. **(10 marks)**

c) Based on the scenario, identify **MMRA**'s likely current maturity level and justify your assessment. **(04 marks)**

d) Outline a step-by-step plan **MMRA** could follow to move from its current level to Level **4** or Level **5**. **(06 marks)**

e) Discuss the benefits and challenges of applying the CMM in a public sector regulatory environment in Zambia. **(04 marks)**

f) Propose **(03) THREE** key performance indicators (KPIs) **MMRA** could use to monitor cybersecurity maturity progress. **(06 marks)**

g) Distinguish between **CMM** and **CMMI**. **(04 marks)**

(Total marks for Section A: 40 marks)

SECTION B

CHOOSE ANY THREE (03) QUESTIONS FROM THIS SECTION

QUESTION TWO

a) Explain how exploratory testing can be applied to ensure the reliability and security of mobile money services, such as those offered by Mobile Money Operators? **(04 marks)**

b) Describe the role that exploratory testing play in identifying usability issues and improving the overall user experience? **(04 marks)**

c) You are tasked with testing the login functionality of a school web application. Describe the critical test scenarios you would cover. **(08 marks)**

d) Describe **TWO (02)** Quality Assurance Software testing tools. **(04 marks)**

(Total: 20 marks)

QUESTION THREE

CASE STORY:

A new company, **LamyaNet** has been providing mobile network services in Lusaka. They recently deployed a new network infrastructure to upgrade from 5G to 6G, but during maintenance, the Core Network Department has reported latency, packet loss, and inconsistent speeds. You have just joined **LamyaNet** as a Software Engineer under the Core Network Department, explain how you will;

a) Test, **(05 marks)**

b) Analyze, and **(05 marks)**

c) Optimize the network for better performance and reliability. **(05 marks)**

d) Explain the concept of **testing the software blindfolded** in dynamic testing. **(05 marks)**

(Total: 20 Marks)

QUESTION FOUR

a) Describe how Artificial Intelligence (AI) and Machine Learning (ML) testing can be applied to ensure the reliability and accuracy of AI-powered agricultural systems, such as those used for crop yield prediction or disease detection? **(08 marks)**

b) Explain black box testing and white box testing. **(06 marks)**

c) Distinguish between static testing and dynamic testing. **(04 marks)**

d) Describe a bug in software development. **(02 marks)**

(Total: 20 marks)

QUESTION FIVE

a) Describe how bugs impact software quality and user experience. **(04 marks)**

b) Give **(04) FOUR** reasons whether all software bugs can be eliminated or not. **(04 marks)**

c) Describe Bug Bashes and Beta Testing and the role they play in Software Quality Assurance and Testing. **(04 marks)**

d) Explain testing specification in static testing. **(02 marks)**

e) Describe the various specialized techniques that are used in Software testing.

(06 marks)

(Total: 20 marks)

END OF EXAMINATION PAPER